```python
transactions = [
    (101, 500, "Deposit"),
    (102, 200, "Withdraw"),
    (101, 300, "Deposit"),
    (103, -100, "Withdraw"),
    (104, 700, "Deposit")
]

seen = []

for tid, amount, ttype in transactions:
    if tid in seen:
        print(tid, "Duplicate transaction ID")
    else:
        seen.append(tid)

    if amount <= 0:
        print(tid, "Invalid amount")

print("Suspicious sequences:")

for i in range(1, len(transactions)):
    if transactions[i][2] == transactions[i-1][2]:
```

Problems Output Debug Console Terminal Ports

Python

```
PS C:\Users\HP\AppData\Local\Programs\Microsoft VS Code> & C:\Users\HP\AppData\Local\Programs\Python\Python311\python.exe c:/Users/HP/Desktop/python-abl/task1.
py
101 Duplicate transaction ID
103 Invalid amount
Suspicious sequences:
PS C:\Users\HP\AppData\Local\Programs\Microsoft VS Code>
```

C: > Users > HP > Desktop > python-abl > task2.py > scores

```python
scores = [80, 60, 90, 70, 85, 65]
group_size = 2

scores.sort(reverse=True)
groups = [[] for i in range(len(scores) // group_size)]
totals = [0] * len(groups)

for score in scores:
    index = totals.index(min(totals))
    if len(groups[index]) < group_size:
        groups[index].append(score)
        totals[index] += score
    else:
        for i in range(len(groups)):
            if len(groups[i]) < group_size:
                groups[i].append(score)
                totals[i] += score
                break

for i in range(len(groups)):
    print("Group", i + 1, ":", groups[i])
    print("Total score:", totals[i])
```

Problems Output Debug Console Terminal Ports Python

```
Group 3 : [80, 70]
Total score: 150
PS C:\Users\HP\AppData\Local\Programs\Microsoft VS Code> & C:\Users\HP\AppData\Local\Programs\Python\Python311\python.exe c:/Users/HP/Desktop/python-abl/task2.
py
Group 1 : [90, 60]
Total score: 150
Group 2 : [85, 65]
Total score: 150
Group 3 : [80, 70]
Total score: 150
PS C:\Users\HP\AppData\Local\Programs\Microsoft VS Code>
```

```python
stock = {"Burger": 10, "Pizza": 5, "Juice": 8}
price = {"Burger": 50, "Pizza": 100, "Juice": 30}
sales = 0

orders = [
    ("Burger", 2),
    ("Pizza", 1),
    ("Juice", 3),
    ("Pizza", 6)
]

for item, qty in orders:
    if item in stock and qty > 0 and qty <= stock[item]:
        bill = price[item] * qty
        stock[item] -= qty
        sales += bill
        print(item, "Bill:", bill)
    else:
        print(item, "Order rejected")

print("Total sales:", sales)
print("Remaining stock:", stock)
```

Problems Output Debug Console Terminal Ports Python

```
Total sales: 290
Remaining stock: {'Burger': 8, 'Pizza': 4, 'Juice': 5}
PS C:\Users\HP\AppData\Local\Programs\Microsoft VS Code> & C:\Users\HP\AppData\Local\Programs\Python\Python311\python.exe c:/Users/HP/Desktop/python-abl/task3.
py
Burger Bill: 100
Pizza Bill: 100
Juice Bill: 90
Pizza Order rejected
Total sales: 290
Remaining stock: {'Burger': 8, 'Pizza': 4, 'Juice': 5}
PS C:\Users\HP\AppData\Local\Programs\Microsoft VS Code>
```

```python
books = [
    ("B101", 5, 8),
    ("B102", 7, 7),
    ("B103", 4, 10),
    ("B104", 6, 9)
]

available = 4S
fine_per_day = 5
total_fine = 0

for book, allowed, returned in books:
    delay = returned - allowed

    if delay > 0:
        fine = delay * fine_per_day
        total_fine += fine
        print(book, "Overdue days:", delay, "Fine:", fine)
    else:
        print(book, "Returned on time")

    available += 1
```

Problems 2 Output Debug Console Terminal Ports Python

```
Total fine: 60
Available books: 8
PS C:\Users\HP\AppData\Local\Programs\Microsoft VS Code> & C:\Users\HP\AppData\Local\Programs\Python\Python311\python.exe c:/Users/HP/Desktop/python-abl/task4.
py
B101 Overdue days: 3 Fine: 15
B102 Returned on time
B103 Overdue days: 6 Fine: 30
B104 Overdue days: 3 Fine: 15
Total fine: 60
Available books: 8
PS C:\Users\HP\AppData\Local\Programs\Microsoft VS Code>
```

```python
routes = [("Route A", 120), ("Route B", 85), ("Route C", 200)]
capacity = 50
total_buses = 0
total_unserved = 0

for route, passengers in routes:
    buses = passengers // capacity

    if passengers % capacity != 0:
        buses += 1

    served = buses * capacity
    unserved = max(0, passengers - served)

    total_buses += buses
    total_unserved += unserved

    print(route, "Buses needed:", buses)
    print("Unserved passengers:", unserved)

print("Total buses:", total_buses)
print("Total unserved:", total_unserved)
```

Problems 2 Output Debug Console Terminal Ports Python

```
PS C:\Users\HP\AppData\Local\Programs\Microsoft VS Code> & C:\Users\HP\AppData\Local\Programs\Python\Python311\python.exe c:/Users/HP/Desktop/python-abl/task5.
py
Route A Buses needed: 3
Unserved passengers: 0
Route B Buses needed: 2
Unserved passengers: 0
Route C Buses needed: 4
Unserved passengers: 0
Total buses: 9
Total unserved: 0
PS C:\Users\HP\AppData\Local\Programs\Microsoft VS Code>
```

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```python
products = ["Product A", "Product B", "Product C"]

production = [
    [100, 200, 150],
    [5, 30, 10]
]

threshold = 10

for i in range(len(products)):
    total = production[0][i]
    defective = production[1][i]
    percentage = defective / total * 100

    print(products[i])
    print("Production:", total)
    print("Defective units:", defective)
    print("Defect percentage:", round(percentage, 2), "%")

    if percentage > threshold:
        print("Quality threshold exceeded")
    else:
        print("Quality acceptable")
```

Problems 2 | Output | Debug Console | Terminal | Ports

Python

```
Product B
Production: 200
Defective units: 30
Defect percentage: 15.0 %
Quality threshold exceeded
Product C
Production: 150
Defective units: 10
Defect percentage: 6.67 %
Quality acceptable
PS C:\Users\HP\AppData\Local\Programs\Microsoft VS Code>
```

```python
seats = [
    [101, 102, 0],
    [103, 104, 105],
    [101, -1, 106]
]

reserved = [-1]
roll_numbers = []
empty = 0
invalid = 0

for row in seats:
    for seat in row:
        if seat == 0:
            empty += 1
        elif seat == -1:
            print("Reserved seat")
        elif seat < -1:
            invalid += 1
            print("Invalid allocation:", seat)
        elif seat in roll_numbers:
            invalid += 1
            print("Duplicate roll number:", seat)
```

Problems 2 | Output | Debug Console | Terminal | Ports

```
Empty seats: 1
Invalid allocations: 1
Valid students: 6
PS C:\Users\HP\AppData\Local\Programs\Microsoft VS Code> & C:\Users\HP\AppData\Local\Programs\Python\Python311\python.exe c:/Users/HP/Desktop/python-abl/task7.
py
Duplicate roll number: 101
Reserved seat
Empty seats: 1
Invalid allocations: 1
Valid students: 6
PS C:\Users\HP\AppData\Local\Programs\Microsoft VS Code>
```

```python
areas = ["Area A", "Area B", "Area C"]

demand = [
    [100, 120, 110],
    [150, 130, 140],
    [90, 100, 95]
]

supply = [
    [90, 100, 100],
    [120, 100, 120],
    [90, 90, 80]
]

shortages = []

for i in range(len(areas)):
    total_demand = sum(demand[i])
    total_supply = sum(supply[i])
    shortage = total_demand - total_supply

    if shortage < 0:
        shortage = 0
```

Problems 2 | Output | Debug Console | Terminal | Ports

Python

```
Shortage: 40
Area B
Demand: 420
Supply: 340
Shortage: 80
Area C
Demand: 285
Supply: 260
Shortage: 25
Most underserved area: Area B
PS C:\Users\HP\AppData\Local\Programs\Microsoft VS Code>
```

Ln 1, Col 1 | Spaces: 4 | UTF-8 | CRLF | { } Python | Python 3.11 (64-bit)

Search

ENG IN

15:42

```python
products = ["Pen", "Book", "Bag"]

prices = [
    [10, 50, 500],
    [12, 45, 600],
    [15, 60, 550]
]

threshold = 20
fluctuations = []

for i in range(len(products)):
    old_price = prices[0][i]
    new_price = prices[-1][i]

    change = (new_price - old_price) / old_price * 100
    fluctuations.append(abs(change))

    print(products[i])
    print("Price change:", round(change, 2), "%")

    if abs(change) > threshold:
        print("Threshold exceeded")
```

Problems 2 | Output | Debug Console | Terminal | Ports

```
Pen
Price change: 50.0 %
Threshold exceeded
Book
Price change: 20.0 %
Within threshold
Bag
Price change: 10.0 %
Within threshold
Maximum fluctuation: Pen
PS C:\Users\HP\AppData\Local\Programs\Microsoft VS Code>
```

```python
registrations = [
    ("S101", "Python"),
    ("S102", "Java"),
    ("S101", "Python"),
    ("S103", "Advanced Python"),
    ("S104", "Java")
]

valid_courses = ["Python", "Java", "Math"]
seen = []

for student, course in registrations:
    record = (student, course)

    if record in seen:
        print("Duplicate registration:", record)
    else:
        seen.append(record)

    if course not in valid_courses:
        print("Invalid course:", student, course)

print("Valid unique registrations:")
```

Problems 2 | Output | Debug Console | Terminal | Ports

```
S102 Java
S104 Java
PS C:\Users\HP\AppData\Local\Programs\Microsoft VS Code> & C:\Users\HP\AppData\Local\Programs\Python\Python311\python.exe c:/Users/HP/Desktop/python-abl/task10
.py
Duplicate registration: ('S101', 'Python')
Invalid course: S103 Advanced Python
Valid unique registrations:
S101 Python
S102 Java
S104 Java
PS C:\Users\HP\AppData\Local\Programs\Microsoft VS Code>
```

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Search

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